Venkata Lakshmi Parimala Pasupuleti

Software Engineer · Python · TypeScript · Node.js · React · Cloud-Native · REST APIs · AWS · Agentic Workflows · Real Estate Domain

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PROFESSIONAL SUMMARY

Full-stack Software Engineer with 8+ years building cloud-native backend services and scalable web applications in real estate software, with a track record of platform modernization — transforming legacy workflows into modern, extensible services designed for long-term evolution and AI integration. At Entrata (PropTech / Real Estate SaaS), owned 20+ production modules end-to-end — Python, TypeScript, Node.js, React, Angular, PostgreSQL, NoSQL, AWS — for 100K+ monthly active users at 99.5% uptime. Delivered legacy modernization at scale: RabbitMQ async redesign, Docker microservices (50% scalability improvement), AWS S3 migrations. AI-ready by default: built AgentProbe (open-source eval toolkit) and Leasel (8-agent LangGraph agentic workflow platform, 93.6% precision) — the agentic workflow and AI adoption foundations Genesis Capital is building toward. Uses AI-assisted development tools daily. M.S. Data Analytics Engineering, George Mason University (GPA 3.87). — github.com/venkatalakshmiparimala/AgentProbe

WORK EXPERIENCE

Blue Horse Digital
Sep 2026 – Present

AI/ML Engineer (Volunteer) · Remote

- Building ML models, data pipelines, and production APIs; integrating AI components across backend and reporting layers; troubleshooting and resolving defects across the application stack.

Magizhs Tech Solution
May 2025 – Aug 2025

AI/ML Engineer Intern · Remote

- Built cloud-native agentic pipeline (LangGraph) translating natural-language requests into structured outputs; designed RESTful API and integration layers for automation consumption; engineered validation guardrails; continuously evaluated emerging AI tools for practical adoption.

George Mason University
Aug 2024 – Aug 2025

Research Associate + Teaching Assistant

- Built multi-source ML prediction pipelines (Python, R) across 190+ countries (R^2=0.977); HuggingFace fine-tuning + inference API deployment; statistical modeling and validation. Workshops to 600+ students across Python and SQL.

Entrata (PropTech / Real Estate SaaS)
Sep 2018 – Aug 2024

Senior Software Engineer · Remote

- **Full-stack development across 20+ production modules** (Python, TypeScript, Node.js, React, Angular, PostgreSQL, NoSQL, RabbitMQ, AWS) serving 100K+ monthly active users at 99.5% uptime — designed, built, tested, deployed, and operated end-to-end across 120+ Agile sprints. Translated product requirements and wireframes into high-quality, maintainable code. Took ownership and accountability for delivered solutions with minimal supervision. Zero compliance violations. Zero security incidents.
- **Platform modernization and scalable system design** — led AWS S3 document migration (100K+ units, 100% PHI compliance); redesigned legacy RabbitMQ async processing with retry logic and idempotency guarantees; Docker microservices redesign (50% scalability improvement, 40% deployment time reduction); 200+ SQL optimizations (30% performance gain). Identified bottlenecks, troubleshot production issues, and resolved defects across the full application stack. Designed APIs and services intended for automation and AI consumption.
- **CI/CD, deployment automation, and infrastructure best practices** — CI/CD pipelines (GitHub Actions) with automated deployment workflows, automated testing, and infrastructure-as-code (Terraform); RESTful API and integration layers for partner and third-party systems; OpenTelemetry + Prometheus + Grafana observability (40% debugging time reduction); Python/Bash automation eliminating 80% of manual operational workload. Evaluated emerging technologies for practical adoption. Mentored 5+ engineers; 300+ code reviews.

Entrata India, Pune
Jul 2017 – Sep 2018

Associate Software Engineer

- Built real estate backend modules (PHP, Python, Java, PostgreSQL); led AWS S3 migration; data pipelines across hundreds of thousands of property records; 56 state-specific document templates.

PROJECTS

AgentProbe: Open-Source AI Evaluation Infrastructure
CI-INTEGRATED OPEN SOURCE

Python · FastAPI · LangGraph · Node.js · ChromaDB · GitHub Actions | github.com/venkatalakshmiparimala/AgentProbe

- Cloud-native AI evaluation platform: 15-endpoint FastAPI backend, 5-metric eval engine, LLM-as-judge scoring, CI regression harness with deployment automation, distributed model routing across OpenAI/Anthropic/Groq/Ollama. Designed as an AI-ready, automation-consumable REST API service aligned with infrastructure best practices.

Leasel: Agentic Workflow Platform for Real Estate
AGENTIC WORKFLOWS REAL ESTATE AI

Python · LangGraph · FastAPI · TypeScript-compatible REST APIs · SQLite · OpenAI API | github.com/venkatalakshmiparimala/Lease_complaince_rag

- 8-agent LangGraph agentic workflow platform with durable state management, tool orchestration, RAG over real estate document corpus, and fault-tolerant HITL escalation — built as a scalable service foundation for automation. 93.6% precision and 87.9% F1 across 1,240 real estate compliance test cases.

EDUCATION

M.S. Data Analytics Engineering George Mason University, Virginia — May 2026

GPA: 3.87

B.Tech Computer Science & Engineering JNTU Anantapur — 2017

SKILLS

Languages & Frameworks: Python (8+ yrs), TypeScript, JavaScript, Node.js, Java (Spring Boot), PHP, SQL, PL/SQL, Bash — React.js, Angular, FastAPI, Django, REST APIs, GraphQL, OpenAPI, modern JavaScript frameworks

Backend, Cloud & Infrastructure: PostgreSQL, NoSQL (Redis, MongoDB), RabbitMQ, Apache Kafka, Temporal, Microservices, Cloud-Native Architecture, Service-Oriented Architecture, Distributed Systems, Scalability and Reliability, Multi-Tenant Architecture, Idempotency, Platform Modernization, Legacy Modernization, Third-Party Integrations, Integration Layers, Docker, Kubernetes (K8s), Terraform (IaC), AWS (S3, EC2, RDS, Lambda, Step Functions, Bedrock), Azure, GitHub Actions CI/CD, Deployment Automation, Automated Testing, Unit Testing, Integration Testing

AI & Observability: LangGraph, LangChain, Agentic Workflows, AI-Ready Architecture, AI-Assisted Development, RAG Pipelines, LLM Evaluation, AgentProbe, Document Intelligence, OpenTelemetry, Prometheus, Grafana, PyTorch, HuggingFace, Agile/Scrum, System Design